arXiv:2512.18634v1 [cs.LG] 21 Dec 2025

From Shortcut to Induction Head: How Data Diversity Shapes Algorithm Selection in Transformers

Ryotaro Kawata*[,1,2], Yujin Song*[,1,2], Alberto Bietti[3], Naoki Nishikawa[1,2], Taiji Suzuki[1,2], Samuel Vaiter[4,5], Denny Wu[3,6]

[1]The University of Tokyo, [2]RIKEN AIP, [3]Flatiron Institute, [4]CNRS, [5]Université Côte d'Azur, Laboratoire J.A. Dieudonné, [6]New York University

{kawata-ryotaro725,nishikawa-naoki259}@g.ecc.u-tokyo.ac.jp; y.song.research@gmail.com; {abietti,dwu}@flatironinstitute.org; taiji@mist.i.u-tokyo.ac.jp; samuel.vaiter@cnrs.fr

Abstract

Transformers can implement both generalizable algorithms (e.g., induction heads) and simple positional shortcuts (e.g., memorizing fixed output positions). In this work, we study how the choice of pretraining data distribution steers a shallow transformer toward one behavior or the other. Focusing on a minimal trigger-output prediction task – copying the token immediately following a special trigger upon its second occurrence – we present a rigorous analysis of gradient-based training of a single-layer transformer. In both the infinite and finite sample regimes, we prove a transition in the learned mechanism: if input sequences exhibit sufficient diversity, measured by a low "max-sum" ratio of trigger-to-trigger distances, the trained model implements an induction head and generalizes to unseen contexts; by contrast, when this ratio is large, the model resorts to a positional shortcut and fails to generalize out-of-distribution (OOD). We also reveal a trade-off between the pretraining context length and OOD generalization, and derive the optimal pretraining distribution that minimizes computational cost per sample. Finally, we validate our theoretical predictions with controlled synthetic experiments, demonstrating that broadening context distributions robustly induces induction heads and enables OOD generalization. Our results shed light on the algorithmic biases of pretrained transformers and offer conceptual guidelines for data-driven control of their learned behaviors.

1 Introduction

Large language models (LLMs) leverage circuits of attention heads [VSP+17] to perform (implicit) algorithmic reasoning. Certain attention heads implement discrete algorithms — notably *induction heads* [ENO+21, OEN+22], which scan for previously seen token patterns in the context to predict subsequent tokens. Such heads enable *in-context learning* behaviors [BMR+20], allowing a transformer to continue a sequence such as $[A, B, \ldots, A] \to B$ purely by leveraging patterns in the context. By contrast, attention can also implement positional mechanisms that select tokens based solely on their location in the sequence [VTM+19, AWKA24]. These mechanisms can yield contrasting generalization performance [CBKZ24], and we expect the pretraining data distribution to play a central role in determining which mechanisms a model learns to rely on: depending on structural properties of the corpus, a transformer may either discover generalizable strategies (content-based retrieval) or adopt position-based shortcuts.

Motivation. We theoretically study how pretraining data influences the implemented circuit and out-of-distribution (OOD) generalization performance of the transformer. This perspective is motivated from the empirical observation that pretrained models often leverage shortcut solutions that are brittle beyond the

*Equal contribution.

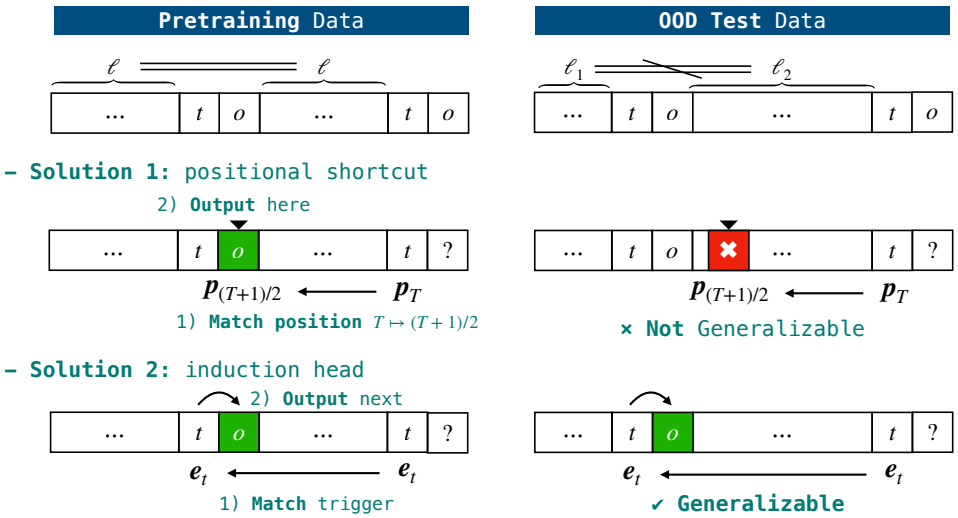

Figure 1: Two mechanisms for the associative copying task `[..., t, o, ..., t]` $\mapsto$ `o`. In the pretraining data, the size of irrelevant tokens before the occurrence of the first and second trigger ℓ remains fixed per sequence, hence allowing two solutions: (*i*) *positional shortcut* that outputs the token at position $(T+1)/2$ for input length T; and (*ii*) *induction head* using token embedding $\boldsymbol{e}$, which finds the queried token and returns the ensuing token. Whereas on OOD sequences with varying $\ell_1 \neq \ell_2$, only (*ii*) remains a valid solution.

training distribution [MPL19, GJM+20, LAG+22]. For instance, a transformer might utilize the aforementioned position-based attention head to memorize that a certain output tends to occur at a particular position in the training text, instead of learning the underlying association (induction head); such positional shortcut is a double-edge sword in algorithmic tasks: transformers can achieve near-perfect accuracy in distribution, but struggle on test sequences of unseen lengths or structures. Since it is empirically known that the learned mechanism heavily depends on the structure of pretraining data [GTLV22, RLIGS22, RPCG23, WNB+25], we ask the following question.

> *How does the data structure decide whether a pretrained transformer implements a generalizable mechanism (e.g.,induction head) or a shortcut that fails OOD (e.g., positional memorization)?*

1.1 Our Contributions

Trigger-output Copying. To investigate this question in a controlled setting, we introduce a minimal *trigger-output copying* task inspired by [BCB+23]. In this synthetic task, each input sequence contains a special `trigger` token that appears twice. The model must predict the token that immediately follows the *first* trigger when the trigger appears the second time. For example, given

$$\ldots[\texttt{trigger}][X]\ldots[\texttt{trigger}][?]\ldots,$$

the correct prediction is X. Depending on the structure of the input sequence, this task admits multiple solutions. We focus on two mechanisms — see Figure 1.

- **Induction head.** The model attends back to the location of the previous trigger and copies the token following it; this works for arbitrarily long gaps between trigger occurrences (up to context-length limit).
- **Positional shortcut.** When the position of the first trigger is inferable from the second (e.g., under periodic structure), the model may copy the token using positional information alone. This shortcut is valid in-distribution but does not reflect the underlying association.

For this task, we define out-of-distribution (OOD) generalization as performance on test sequence with altered structure, where the trigger appears at positions not seen during pretraining (e.g., longer or aperiodic sequences). The induction head mechanism is robust to such shifts as it learns the correct association, whereas the positional shortcut typically fails OOD. Our goal is to identify a data-dependent transition

between these two mechanisms that governs OOD generalization: intuitively, increasing the *diversity* of pretraining sequences — by varying the distances between trigger occurrences — dilutes positional signals and discourages the shortcut; conversely, as the number of trigger tokens in the data grows, the effective signal for induction weakens. We make these intuitions precise in our theoretical analysis.

Main Findings. We provide a quantitative account of how pretraining data diversity shapes the mechanism learned by a pretrained transformer in the trigger-output copying task introduced above. Specifically, we rigorously analyze the in-distribution and out-of-distribution performance of a shallow (single-layer) transformer trained on this synthetic task. By studying an "early-phase" simplification of gradient descent in both the infinite-data (population loss) and finite-data (empirical loss) regimes, we show that the pretraining distribution directly selects the model's algorithm: when pretraining data are sufficiently "diverse" – as measured by a *max-sum ratio* of trigger distances – the transformer learns an induction head; when diversity is low, the model adopts a positional shortcut that fails to generalize OOD. Using this diversity measure that governs the phase transition, we discuss various tradeoffs and how to choose a pretraining distribution that induces the desired induction mechanism with minimal computational cost. Finally, we empirically probe the learned circuits by visualizing attention scores, and present evidence that a similar mechanism transition arises under standard gradient-based training beyond our theoretical setting.

1.2 Related Works

The induction head mechanism in transformers was first presented in the mechanistic interpretability literature [ENO+21, OEN+22], and followup theory investigates when such circuits emerge under simplified training dynamics and tasks [BCB+23, ETE+24, NDL24, Red24, CSWY24]. Empirical studies on algorithmic tasks (copying, arithmetic, sorting) demonstrate that transformers often rely on spurious "shortcut" solutions that fail to generalize, often due to poor use of positional information [ZBB+22, JdDE+23, ZBL+23, GJB+25]; the OOD brittleness of shortcut solution is also documented in [LAG+22]. A complementary thread links the structure of pretraining data to in-context behaviors: the function classes a transformer implements in context and the sensitivity of performance to data statistics such as corpus coverage and frequency [GTLV22, RLIGS22, MLH+22] or task diversity [RPCG23, LLZV+25]. Our analysis aligns with this view by making explicit how diversity in trigger distances steers the learned mechanism. Methodologically, we borrow the "early-phase" simplification of training dynamics and study the loss improvement after the first few gradient descent step [BES+22, DLS22, ORST23, BCB+23].

2 Problem Setting

Notations. For a positive integer N, we denote $[N] := \{1, 2, \ldots, N\}$. For integers $N_1 \le N_2$, we define $[N_1 : N_2] := \{N_1, N_1 + 1, \ldots, N_2\}$. The Softmax function for an N-dimensional vector $\boldsymbol{v} \in \mathbb{R}^N$ is defined as $\mathrm{Softmax}(\boldsymbol{v})_i := \frac{e^{v_i}}{\sum_{j=1}^N e^{v_j}}$. For a vector $\boldsymbol{v}$, we write $\boldsymbol{v} = O_2(f(N))$ if $\|\boldsymbol{v}\|_2 = O(f(N))$, and $\boldsymbol{v} = O_\infty(f(N))$ if $\max_i |v_i| = O(f(N))$. Similar notation is used for a matrix $\boldsymbol{A}$, where $\|\boldsymbol{A}\|_2$ and $\|\boldsymbol{A}\|_\infty$ denote its $\ell_2 \to \ell_2$ spectral and max norms, respectively.

2.1 Data Generating Process

We study the *trigger-output* setting to investigate how transformers acquire the induction head mechanism. Let $N \in \mathbb{N}$ denote the vocabulary size and $L \in \mathbb{N}$ the maximum input sequence length. We designate special tokens as *trigger tokens*. We define our data model as follows:

Definition 1 (Data Distribution)**.** *Let $\ell_1, \ell_2 \in \mathbb{N}$ such that $T := \ell_1 + \ell_2 + 3 \le L - 1$. Let $N_{\mathrm{trg}} \le N$ denote the number of trigger tokens. A sequence $z_{1:T+1} \in [N]^{T+1}$ is sampled as follows:*

1. *Sample a trigger token $t \in [N_{\mathrm{trg}}]$ and an output token $o \in [N_{\mathrm{trg}} + 1 : N]$ uniformly at random, where $N_{\mathrm{trg}} = o(N^{1/3})$.*